Surface Mapping Project: Task Schedule

ECE 467/567 Computer Graphics and CAD, Fall 2026

Revised October 27, 2026, after proposal feedback

Course Deadlines

Date	Deliverable
Sun Oct 18	Proposal submitted (9/10, approved with trimmed core)
Wed Oct 28	Paper choice (individual; Tanner and Drake choose different papers)
Thu Nov 5	Paper presentations
Tue Nov 17	Milestone check
Tue Dec 8	Demo day; final report and code due 11:59 PM

Task Table

Rows are shaded by track: Course & paper, Core build, Stretch, Wrap-up. Stretch tasks can be cut without affecting the core. Overlapping dates on dependent tasks mean the later task can start once the earlier one is underway.

ID	Task	Track	Start	End	Days	Depends on	Owner
P1	Submit proposal (graded 9/10, approved)	Course	Oct 18	Oct 18	1	–	Both
R1	Skim course notes and candidate papers	Course	Oct 19	Oct 27	9	–	Both
P2	Submit paper choices (different papers: LSCM and Sander et al.)	Course	Oct 27	Oct 28	2	R1	Each
P3	Tanner: paper talk and related-works comparison	Course	Oct 29	Nov 4	7	P2	Tanner

ID	Task	Track	Start	End	Days	Depends on	Owner
P4	Drake: paper talk and related-works comparison	Course	Oct 29	Nov 4	7	P2	Drake
S1	Repository and Python environment setup	Core	Oct 19	Oct 21	3	–	
S2	Mesh loading; sphere, hemisphere, torus and disk test meshes	Core	Oct 21	Oct 30	10	S1	
M1	Analytic sphere projections (equirectangular, Mercator, stereographic)	Core	Oct 26	Nov 2	8	S2	
D1	Distortion metrics (per-triangle Jacobian, SVD, fold-overs)	Core	Oct 28	Nov 4	8	S2	
V1	Polyscope viewer: 3D and UV views, method switching	Core	Oct 30	Nov 8	10	S2	
M2	Tutte embedding (uniform and mean-value weights) on disks	Core	Nov 2	Nov 10	9	S2	
D2	Theory tests: Mercator σ_1/σ_2 near 1, Tutte zero fold-overs	Core	Nov 4	Nov 11	8	M1, D1, M2	
V2	Distortion heatmaps and checkerboard texture	Core	Nov 5	Nov 12	8	V1, D1	
M3	LSCM with two pinned boundary vertices	Core	Nov 9	Nov 21	13	M2	
K1	Milestone submission	Course	Nov 16	Nov 17	2	V2, M2, D2	Both
C1	Cut step: Dijkstra seam for closed genus-0 meshes; two cuts for torus	Core	Nov 11	Nov 19	9	M2	
L1	libigl validation checks (not a report result)	Core	Nov 17	Nov 24	8	M2, M3	
E1	Evaluation 1: distortion by method and mesh	Core	Nov 21	Nov 29	9	M3, C1, D2	
E2	Evaluation 2: solve time vs. mesh size	Core	Nov 22	Nov 28	7	M3	
H1	Borrow projector and test hardware	Stretch	Oct 28	Nov 17	21	–	
X1	Cotangent weights, fold-overs reported separately	Stretch	Nov 12	Nov 18	7	M2, D1	
T1	Procedural 3D shaders (moved from core)	Stretch	Nov 18	Nov 25	8	V1	
S3	Phone-scan objects and mesh cleanup (moved from core)	Stretch	Nov 18	Nov 25	8	S2	

ID	Task	Track	Start	End	Days	Depends on	Owner
A1	A1: project onto a known shape	Stretch	Nov 18	Nov 26	9	H1, V1	
B1	ARAP parameterization	Stretch	Nov 22	Dec 1	10	M3	
A2	A2: project onto a scanned object	Stretch	Nov 25	Dec 2	8	A1, S3, C1, M3	
A3	A3: structured-light scanning	Stretch	Nov 30	Dec 5	6	A2	
B2	Multi-chart atlas with chart segmentation	Stretch	Nov 28	Dec 5	8	C1, M3	
F3	Write final report	Wrap-up	Nov 28	Dec 7	10	E1, E2	
F1	Polish viewer and demo script	Wrap-up	Nov 30	Dec 5	6	E1	
F2	Record backup demo video	Wrap-up	Dec 4	Dec 6	3	F1	
F4	Demo day; submit report and code	Wrap-up	Dec 8	Dec 8	1	F2, F3	Both

Changes from the Oct 18 plan: the core now uses one stack (Python and Polyscope). Procedural shaders (T1) and scanned-mesh cleanup (S3) moved to stretch; a cut step (C1) was added; evaluation was reduced to two studies (E1, E2); libigl is used for validation only (L1). Thanksgiving (Nov 26) falls within the evaluation and stretch-goal window.